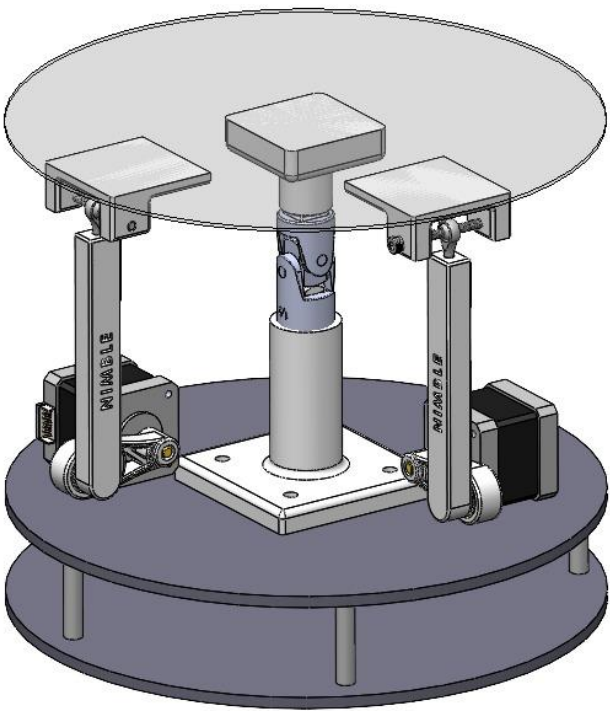


NIMBLE: Computer Vision Based Ball Balancing & Trajectory Tracking Robot

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Overview

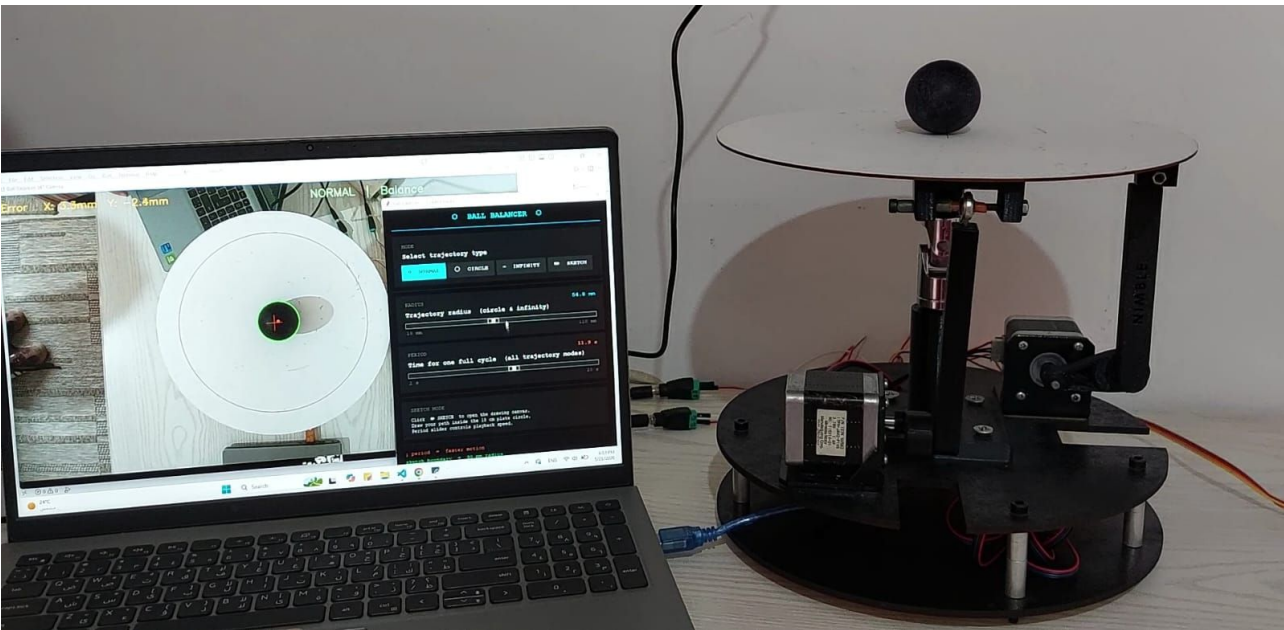
NIMBLE is a vision-based ball balancing robot that keeps a ball at the center of a 2-DOF tilting platform using camera feedback and closed-loop control. The system combines computer vision, embedded control, inverse kinematics and mechanical design to continuously detect the ball's position and adjust the platform angle to stabilize it. In addition to stabilization, the system also supports trajectory tracking, where the ball can be guided along predefined paths, such as a circular/infinity sign trajectory or a custom freehand sketch.



Robot Assembly

Methodology

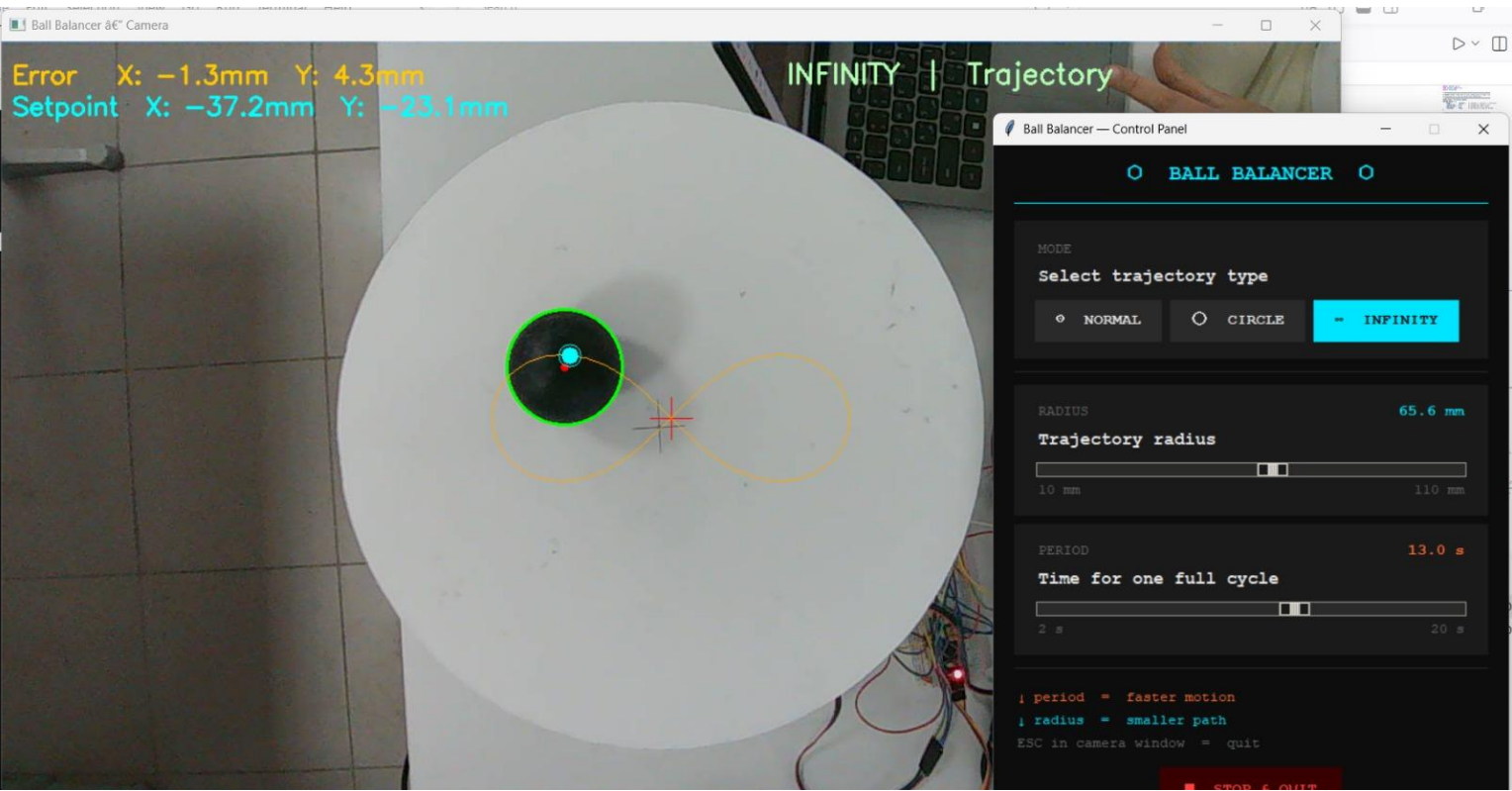
An overhead webcam captures the ball's positional error using OpenCV image processing and transmits it to an STM32 microcontroller. The STM32 runs two independent PID control loops — one per axis — calculating the required tilt angles. These angles are executed by two NEMA 17 stepper motors, which physically tilt the plate and close the feedback loop.



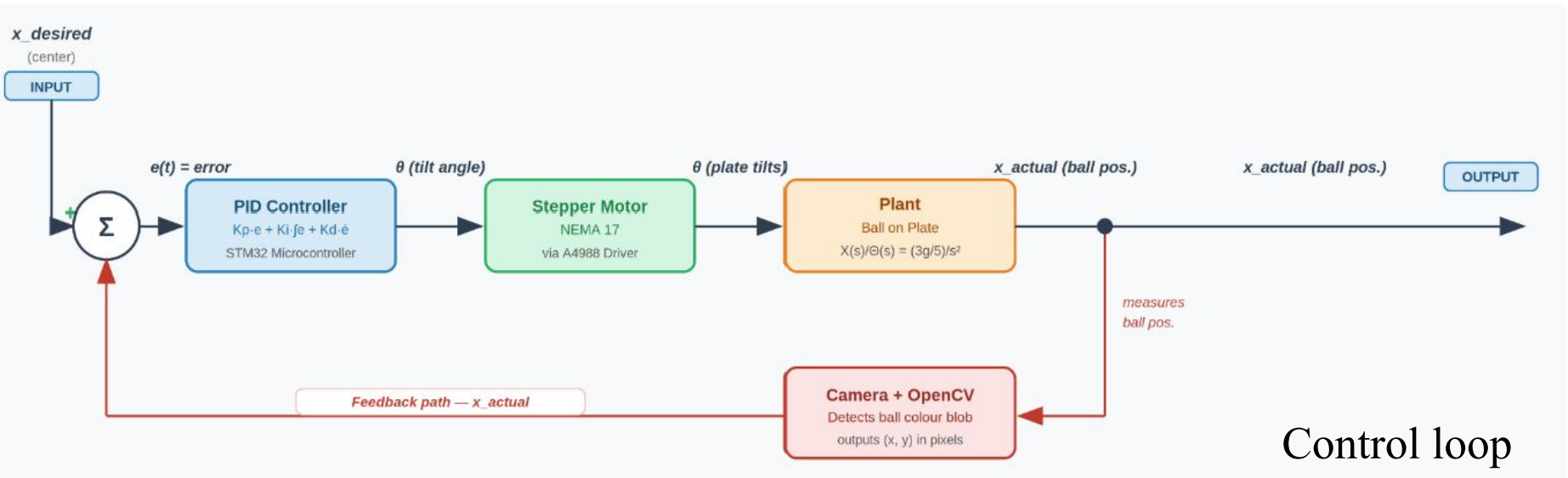
Setup

Computer Vision

Using a 30 fps webcam each frame is processed in Python using OpenCV — converted to HSV color space, thresholded to isolate the black ball, and filtered using circularity and radius checks to reject false detections. An exponential moving average filter then smooths the ball's pixel position frame-to-frame, which is compared against the target setpoint to compute the positional error



Trajectory Tracking



Control loop

Control System Design

The system models a ping-pong ball on a tilted plate as a double integrator plant $G(s) = 5.886/s^2$, making it open-loop unstable. A PID controller stabilizes each axis independently. MATLAB's pidtune achieved $\geq 60^\circ$ phase margin and $\sim 6.78s$ settling time in simulation.